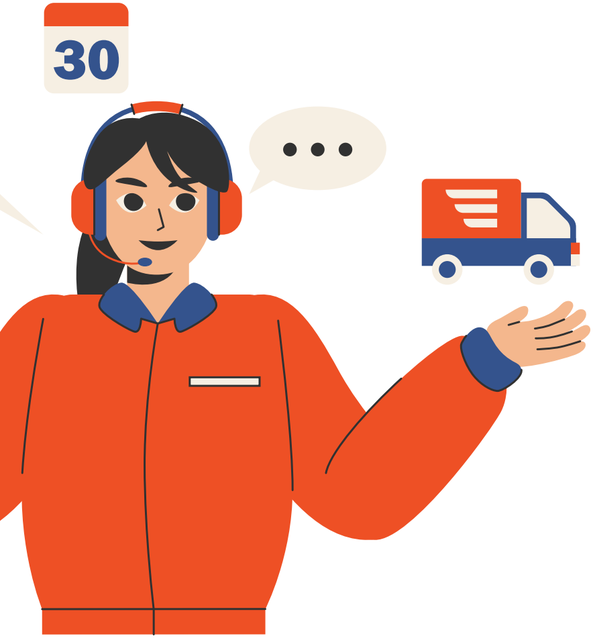
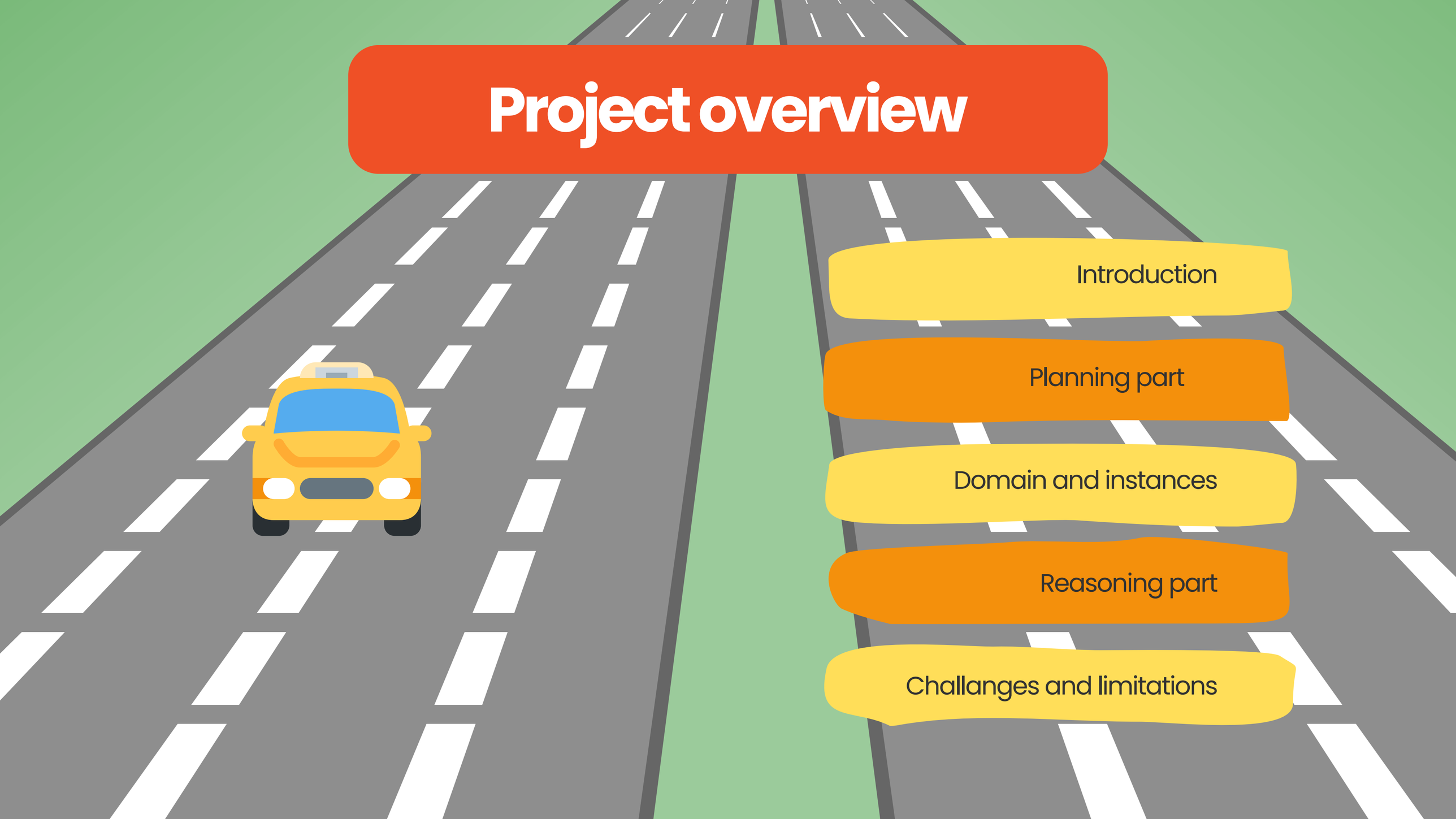



# Intelligent public transport route planning

## with PDDL and IndiGolog

Agnese Chiessi | Ludovica Moscato

# Project overview

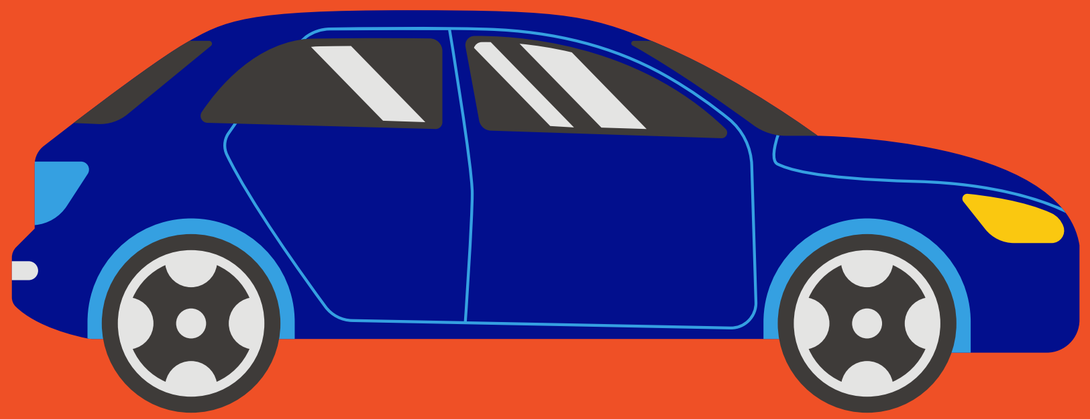


- Introduction
- Planning part
- Domain and instances
- Reasoning part
- Challanges and limitations

ENHSP

# PDDL domain

Goal: Plan optimal routes from one stop to another using public transport (bus, tram, metro), considering time, availability, delays, and user preferences.



# Functions and actions

total-time –> total travel time

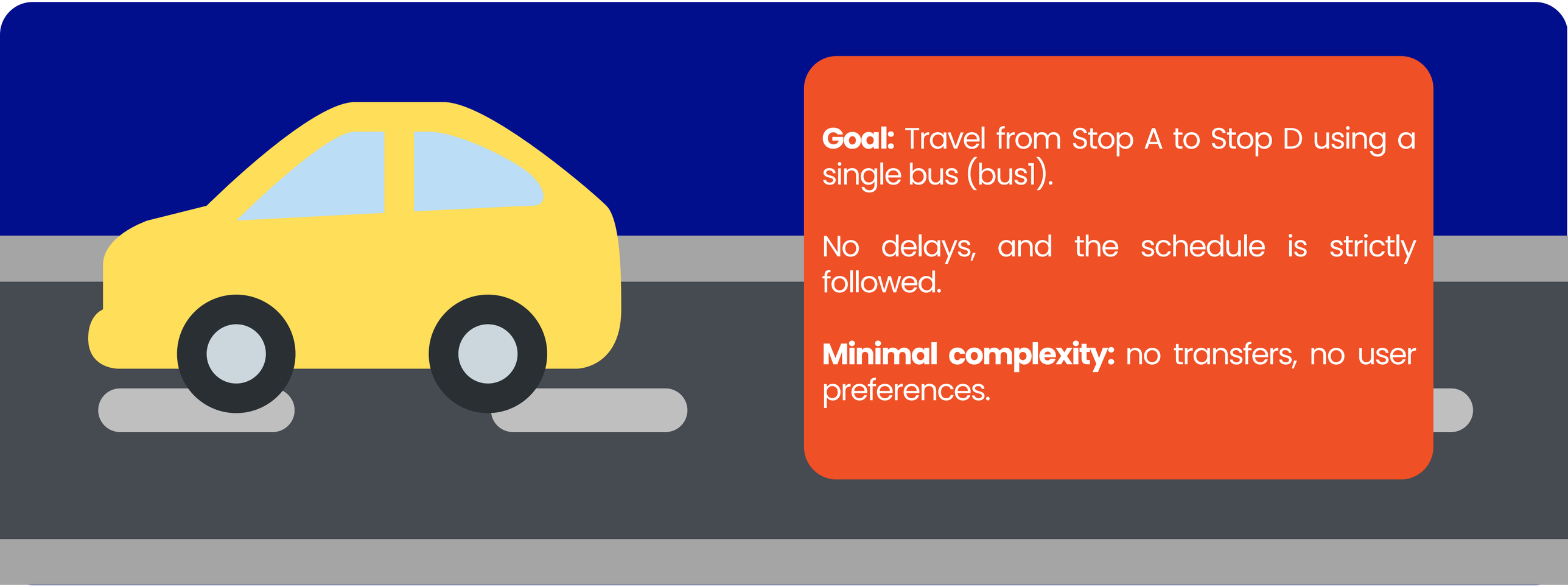
travel-time(v, from, to) –> travel time of the vehicle between stops

departure-time(v, s)–>departure time of the vehicle at a stop

delay(v) –> delay of the vehicle

| Actions | |
|---|---|
| Wait | The person waits 1 time unit |
| Board | The person boards a vehicle at a stop |
| Alight | The person gets off a vehicle at a stop |
| Move-vehicle | The vehicle moves from one stop to another |

# Instance 1: Basic route planning



**Goal:** Travel from Stop A to Stop D using a single bus (bus1).

No delays, and the schedule is strictly followed.

**Minimal complexity:** no transfers, no user preferences.

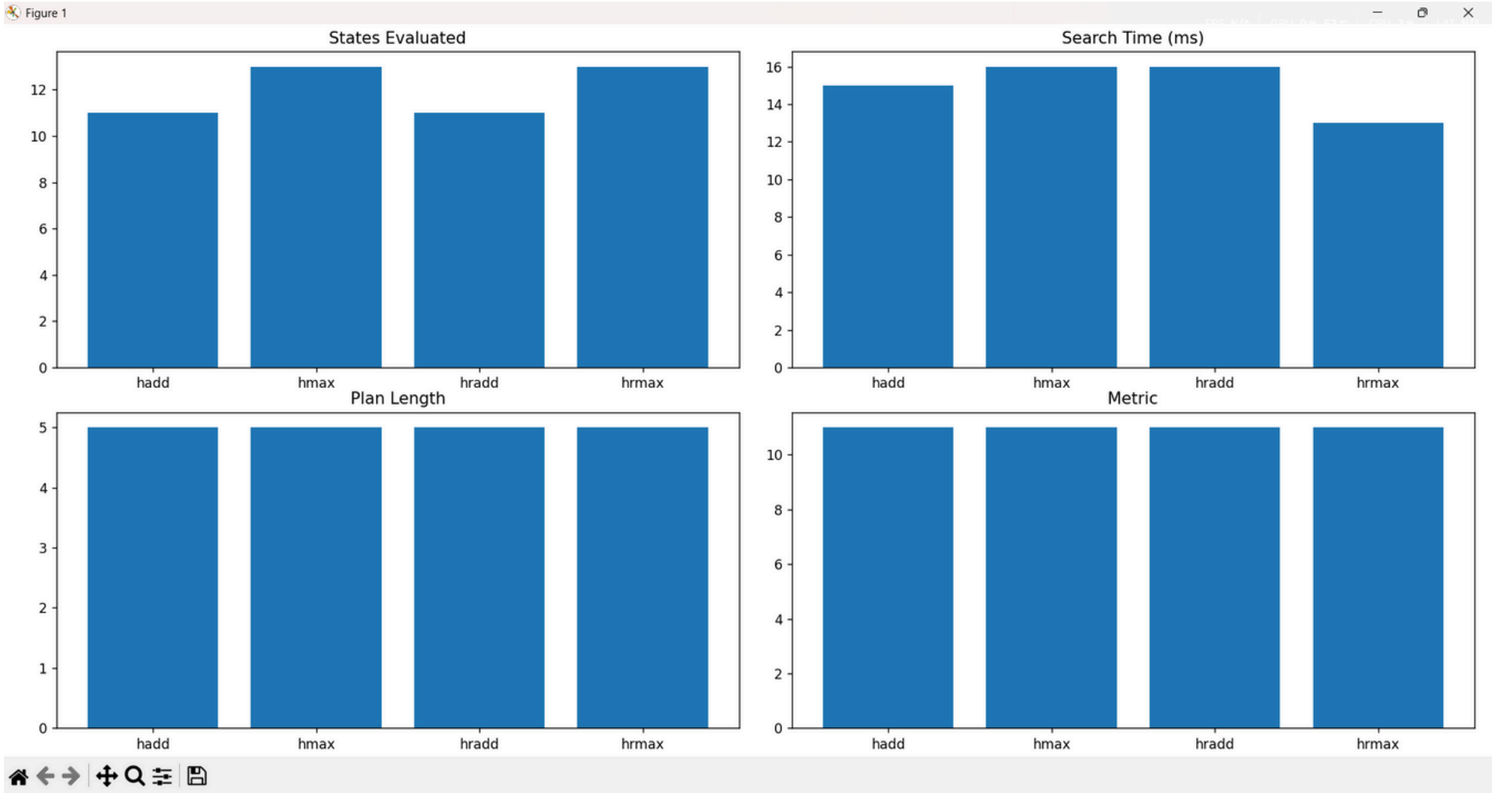
```
(define (problem basic-route)
  (:domain transport)

  (:objects
    user - person
    stop-a stop-b stop-c stop-d - stop
    bus1 - vehicle
  )

  (:init
    (at user stop-a)
    (vehicle-at bus1 stop-a)
    (available bus1)
    (= (departure-time bus1 stop-a) 0)
    (can-travel bus1 stop-a stop-b) (= (travel-time bus1 stop-a stop-b) 5)
    (= (departure-time bus1 stop-b) 5)
    (can-travel bus1 stop-b stop-c) (= (travel-time bus1 stop-b stop-c) 4)
    (= (departure-time bus1 stop-c) 9)
    (can-travel bus1 stop-c stop-d) (= (travel-time bus1 stop-c stop-d) 2)
    (= (delay bus1) 0)

    ; Inizializzazione dei fluenti
    (= (total-time) 0)
  )

  (:goal
    (at user stop-d))
    (:metric minimize (total-time))
)
```

# Instance 1: Init and goal

| Segment | Travel-time | Departure-time |
|---|---|---|
| A->B | 5 min | 0 |
| B->C | 4 min | 5 |
| C->D | 2 min | 9 |

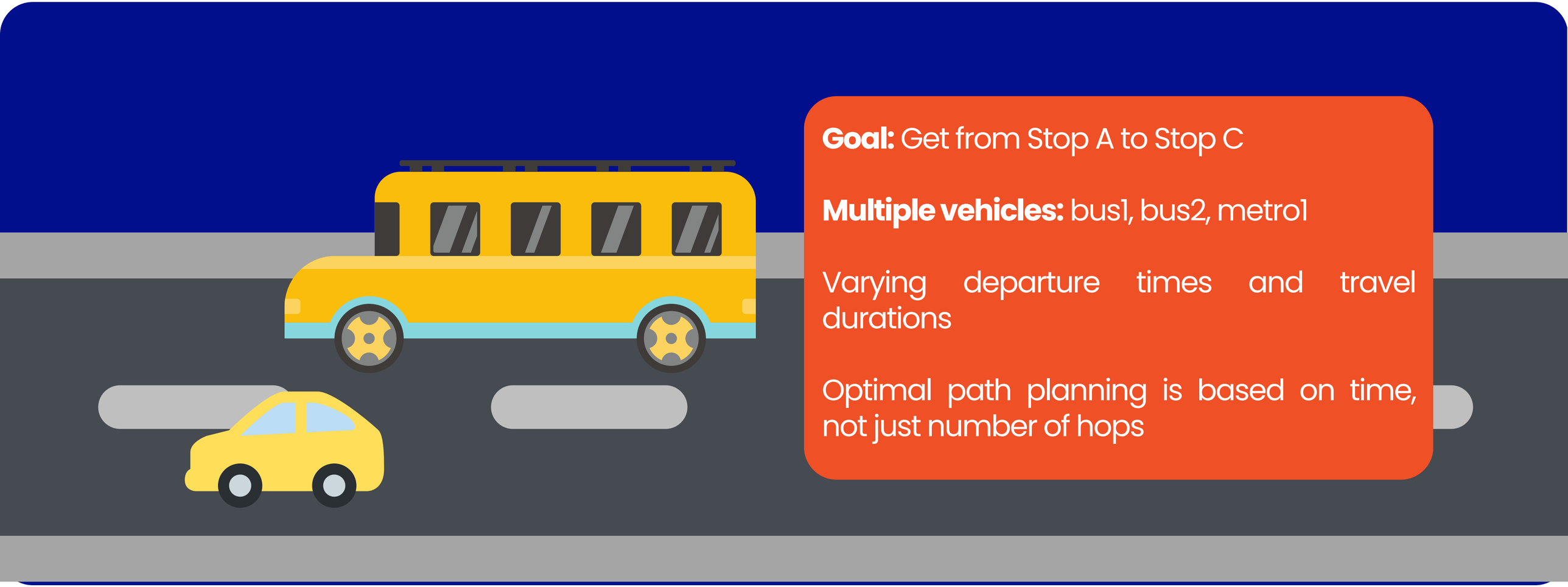
# Instance 1: output

Heuristics used: hmax, hadd, hrmax, hradd
Search algorithms: GBFS, WA*, W_A*_4




```
Domain parsed
Problem parsed
Grounding..
Grounding Time: 20
Aibr Preprocessing
|F|:9
|X|:1
|A|:14
|P|:0
|E|:0
H1 Setup Time (msec): 0
Setting horizon to:NaN
Running Greedy Best First Search
h(n = s_0)=25.0
 g(n)= 5.0 h(n)=10.0
 g(n)= 9.0 h(n)=2.0
 g(n)= 11.0 h(n)=0.0
Problem Solved

Found Plan:
0.0: (board user bus1 stop-a)
1.0: (move-vehicle bus1 stop-a stop-b)
2.0: (move-vehicle bus1 stop-b stop-c)
3.0: (move-vehicle bus1 stop-c stop-d)
4.0: (alight user bus1 stop-d)

Plan-Length:5
Metric (Search):11.0
Planning Time (msec): 166
Heuristic Time (msec): 0
Search Time (msec): 7
```

# Instance 2: Multi route planning



**Goal:** Get from Stop A to Stop C

**Multiple vehicles:** bus1, bus2, metro1

Varying departure times and travel durations

Optimal path planning is based on time, not just number of hops

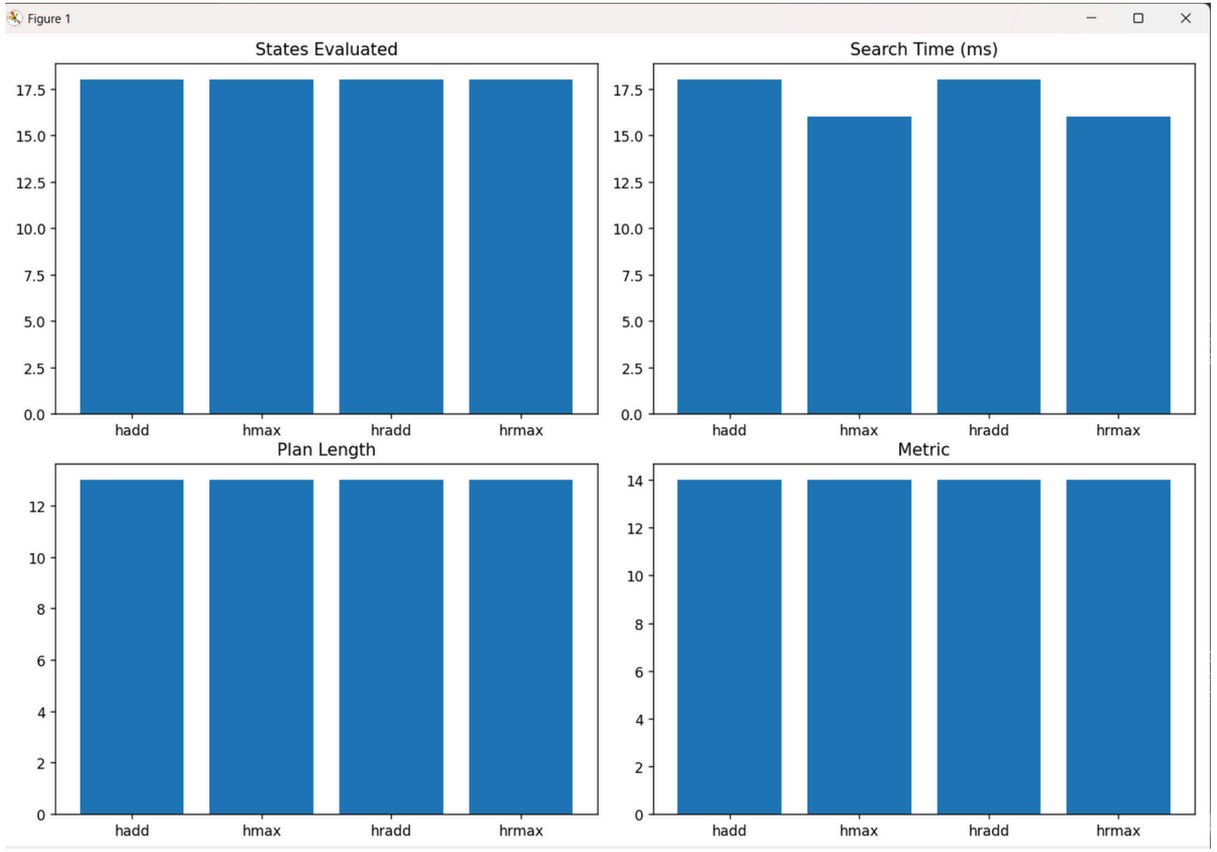
```
(define (problem transport-network)
  (:objects user - person
(:init
  ;; Posizione iniziale utente
  (at user stop-a)

  ;; Posizioni iniziali veicoli
  (vehicle-at bus1 stop-a)
  (vehicle-at bus2 stop-a)
  (vehicle-at metro1 stop-b)
  (available bus1)
  (available bus2)
  (available metro1)

  ;; Connessioni BUS1: A->B->C
  (can-travel bus1 stop-a stop-b)
  (can-travel bus1 stop-b stop-c)
  (= (travel-time bus1 stop-a stop-b) 8)
  (= (travel-time bus1 stop-b stop-c) 10)
  (= (departure-time bus1 stop-a) 30)   ;; spostato molto più tardi
  (= (departure-time bus1 stop-b) 25)   ;; arriva alle 25, parte subito

  ;; Connessioni BUS2: A->D->C
  (can-travel bus2 stop-a stop-d)
  ;(can-travel bus2 stop-d stop-c)
  (= (travel-time bus2 stop-a stop-d) 3)  ;; più veloce da A a D
  (= (travel-time bus2 stop-d stop-c) 4)  ;; più veloce da D a C
  (= (departure-time bus2 stop-a) 5)      ;; parte presto
  (= (departure-time bus2 stop-d) 8)     ;; arriva alle 12, parte subito

  ;; Connessioni METRO1: B->D e D->C (aggiunta D->C)
  (can-travel metro1 stop-b stop-d)
  (can-travel metro1 stop-d stop-c)
  (= (travel-time metro1 stop-b stop-d) 3)
  (= (travel-time metro1 stop-d stop-c) 2) ;; tempo breve da D a C
  (= (departure-time metro1 stop-b) 9)     ;; parte presto da B
  (= (departure-time metro1 stop-d) 12)    ;; parte presto da D

  (= (delay metro1) 0)
  (= (delay bus1) 0)
  (= (delay bus2) 0)
  (= (total-time) 0)
)
  (:goal (at user stop-c))
  (:metric minimize (total-time)))
```

# Instance 2: Init and goal

| Vehicle | Route | Travel-time | Departure-time |
|---|---|---|---|
| Bus1 | A->B->C | A-B: 8;<br>B-C: 10 | A: 30;<br>B: 25 |
| Bus2 | A->D->C | A-D: 3;<br>D-C: 4 | A: 5;<br>D: 8 |
| Metro1 | B->D->C | B-D: 3;<br>D-C: 2 | B: 9;<br>D: 12 |

# Instance 2: Output

Heuristics used: hmax, hadd, hrmax, hradd
Search algorithms: GBFS, WA*, W_A*_4




```
Aibr Preprocessing
|F|:15
|X|:1
|A|:23
|P|:0
|E|:0
H1 Setup Time (msec): 10
Setting horizon to:NaN
Running Greedy Best First Search
h(n = s_0)=63.0
 g(n)= 1.0 h(n)=57.0
 g(n)= 2.0 h(n)=51.0
 g(n)= 3.0 h(n)=45.0
 g(n)= 4.0 h(n)=39.0
 g(n)= 5.0 h(n)=33.0
 g(n)= 8.0 h(n)=18.0
 g(n)= 9.0 h(n)=14.0
 g(n)= 12.0 h(n)=2.0
 g(n)= 14.0 h(n)=0.0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (wait user stop-a)
3.0: (wait user stop-a)
4.0: (wait user stop-a)
5.0: (board user bus2 stop-a)
6.0: (move-vehicle bus2 stop-a stop-d)
7.0: (alight user bus2 stop-d)
8.0: (wait user stop-d)
9.0: (move-vehicle metro1 stop-b stop-d)
10.0: (board user metro1 stop-d)
11.0: (move-vehicle metro1 stop-d stop-c)
12.0: (alight user metro1 stop-c)

Plan-Length:13
Metric (Search):14.0
Planning Time (msec): 171
Heuristic Time (msec): 4
Search Time (msec): 9
```

# Instance 3: Delay-aware route planning

A complex multimodal transport network with trams, metros, and buses

**Goal:** Reach stop-j starting from stop-a, minimizing total travel time

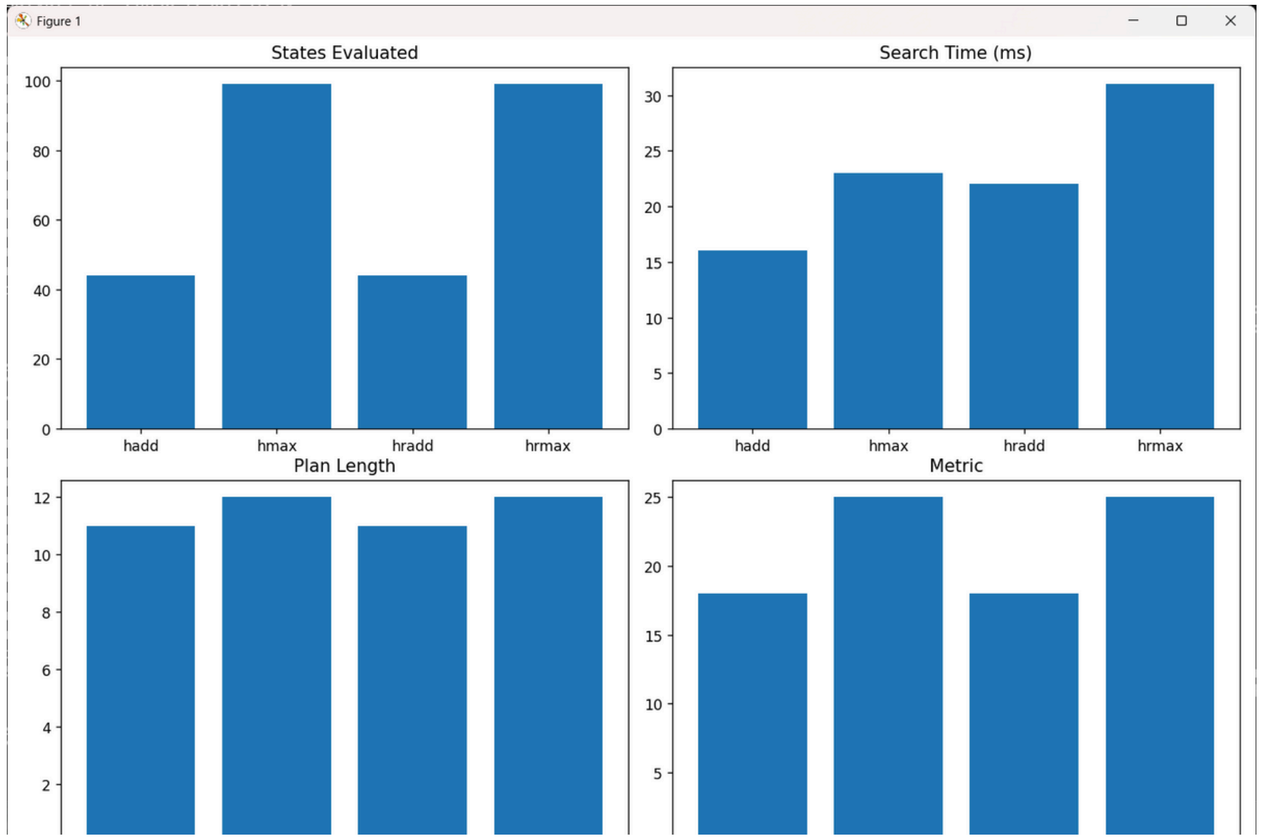
| Vehicle | Route | Travel-time | Departure-time | Delay | Preferences |
|---|---|---|---|---|---|
| Bus1 | B → F → J | B→F: 6, F→J: 3 | B: 7, F: 13 | 1 | Allowed |
| Tram1 | A → B → C → D | A→B: 3, B→C: 3, C→D: 4 | A: 2, B: 5, C: 8 | 0 | Allowed |
| Bus2 | C → F → G → H | F→H: 4, H→I: 2 | C: 9, F: 13, G: 16 | 3 | Allowed |
| Metro1 | B->D->C | C→F: 4, F→G: 3, G→H: 5 | B: 9;<br>D: 12 | 0 | Allowed |
| Metro2 | E → G → I | E→G: 5, G→I: 4 | E: 10, G: 15 | 2 | Allowed |
| Tram2 | A → D → E → F | A→D: 2, D→E: 2, E→F: 3<br>5 | A: 2, D: 4, E: 6 | 5 | Avoided |

# Instance 3: Init and goal

```
(define (problem transport-delay-avoid-complex-v2)
  (:domain transport)
  (:objects
    user - person
    stop-a stop-b stop-c stop-d stop-e stop-f stop-g stop-h stop-i stop-j - stop
    tram1 tram2 metro1 metro2 bus1 bus2 - vehicle
  )
  (:init
    (at user stop-a)
    ;; posizioni iniziali veicoli
    (vehicle-at tram1 stop-a)
    (vehicle-at tram2 stop-a)
    (vehicle-at metro1 stop-c)
    (vehicle-at metro2 stop-e)
    (vehicle-at bus1 stop-b)
    (vehicle-at bus2 stop-f)
    ;; percorsi
    (can-travel tram1 stop-a stop-b)
    (can-travel tram1 stop-b stop-c)
    (can-travel tram1 stop-c stop-d)
    (can-travel tram2 stop-a stop-d)
    (can-travel tram2 stop-d stop-e)
    (can-travel tram2 stop-e stop-f)
    (can-travel metro1 stop-c stop-f)
    (can-travel metro1 stop-f stop-g)
    (can-travel metro1 stop-g stop-h)
    (can-travel metro2 stop-e stop-g)
    (can-travel metro2 stop-g stop-i)
    (can-travel bus1 stop-b stop-f)
    (can-travel bus1 stop-f stop-j)
    (can-travel bus2 stop-f stop-h)
    (can-travel bus2 stop-h stop-i)
    ;; tempi di viaggio (in minuti)
    (= (travel-time tram1 stop-a stop-b) 3)
    (= (travel-time tram1 stop-b stop-c) 3)
    (= (travel-time tram1 stop-c stop-d) 4)
    (= (travel-time tram2 stop-a stop-d) 2)
    (= (travel-time tram2 stop-d stop-e) 2)
    (= (travel-time tram2 stop-e stop-f) 3)
    (= (travel-time metro1 stop-c stop-f) 4)
    (= (travel-time metro1 stop-f stop-g) 3)
    (= (travel-time metro1 stop-g stop-h) 5)
    (= (travel-time metro2 stop-e stop-g) 5)
    (= (travel-time metro2 stop-g stop-i) 4)
    (= (travel-time bus1 stop-b stop-f) 6)
    (= (travel-time bus1 stop-f stop-j) 3)
    (= (travel-time bus2 stop-f stop-h) 4)
    (= (travel-time bus2 stop-h stop-i) 2)
    ;; orari di partenza
    (= (departure-time tram1 stop-a) 2)
    (= (departure-time tram1 stop-b) 5)
    (= (departure-time tram1 stop-c) 8)
    (= (departure-time tram2 stop-a) 2)
    (= (departure-time tram2 stop-d) 4)
    (= (departure-time tram2 stop-e) 6)
    (= (departure-time metro1 stop-c) 9)
    (= (departure-time metro1 stop-f) 13)
    (= (departure-time metro1 stop-g) 16)
    (= (departure-time metro2 stop-e) 10)
    (= (departure-time metro2 stop-g) 15)
    (= (departure-time bus1 stop-b) 7)
    (= (departure-time bus1 stop-f) 13)
    (= (departure-time bus2 stop-f) 14)
    (= (departure-time bus2 stop-h) 18)
    ;; ritardi (delay in minuti)
    (= (delay tram2) 5)
    (= (delay metro2) 2)
    (= (delay tram1) 0)
    (= (delay metro1) 0)
    (= (delay bus1) 1)
    (= (delay bus2) 3)
    ;; disponibilità
    (available tram1)
    (available tram2)
    (available metro1)
    (available metro2)
    (available bus1)
    (available bus2)
    ;; preferenze: evita tram2
    (avoid user tram2)
    ;; inizializzazione fluente
    (= (total-time) 0)
  )
  (:goal (at user stop-j))
  (:metric minimize (total-time))
)
```

# Instance 3: Output

Heuristics used: hmax, hadd, hrmax, hradd
Search algorithms: GBFS, WA*, W_A*_4




```
|F|:15
|X|:1
|A|:23
|P|:0
|E|:0
H1 Setup Time (msec): 16
Setting horizon to:NaN
Running Greedy Best First Search
h(n = s_0)=63.0
 g(n)= 1.0 h(n)=57.0
 g(n)= 2.0 h(n)=51.0
 g(n)= 3.0 h(n)=45.0
 g(n)= 4.0 h(n)=39.0
 g(n)= 5.0 h(n)=33.0
 g(n)= 8.0 h(n)=18.0
 g(n)= 9.0 h(n)=14.0
 g(n)= 12.0 h(n)=2.0
 g(n)= 14.0 h(n)=0.0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (wait user stop-a)
3.0: (wait user stop-a)
4.0: (wait user stop-a)
5.0: (board user bus2 stop-a)
6.0: (move-vehicle bus2 stop-a stop-d)
7.0: (alight user bus2 stop-d)
8.0: (wait user stop-d)
9.0: (move-vehicle metro1 stop-b stop-d)
10.0: (board user metro1 stop-d)
11.0: (move-vehicle metro1 stop-d stop-c)
12.0: (alight user metro1 stop-c)

Plan-Length:13
Metric (Search):14.0
Planning Time (msec): 169
Heuristic Time (msec): 16
```

# Comparison between heuristic **hadd** and all search algorithms on instance 3

```
Grounding..
Grounding Time: 20
Aibr Preprocessing
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 10
Setting horizon to:NaN
Running greedy WA-STAR with hw = 4
h(n = s_0)=45.0
f(n) = 180.0 (Expanded Nodes: 0, Evaluated States: 0, Time: 0.006) Frontier Size: 0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (alight user bus1 stop-j)

Plan-Length:11
Metric (Search):18.0
Planning Time (msec): 223
Heuristic Time (msec): 6
Search Time (msec): 21
Expanded Nodes:12
States Evaluated:38
```

```
Grounding Time: 20
Aibr Preprocessing
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 9
Setting horizon to:NaN
Running Greedy Best First Search
h(n = s_0)=45.0
 g(n)= 1.0 h(n)=40.0
 g(n)= 2.0 h(n)=35.0
 g(n)= 5.0 h(n)=23.0
 g(n)= 6.0 h(n)=20.0
 g(n)= 7.0 h(n)=17.0
 g(n)= 14.0 h(n)=4.0
 g(n)= 18.0 h(n)=0.0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (alight user bus1 stop-j)

Plan-Length:11
Metric (Search):18.0
Planning Time (msec): 211
Heuristic Time (msec): 4
Search Time (msec): 15
Expanded Nodes:13
States Evaluated:44
```

```
Grounding Time: 21
Aibr Preprocessing
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 12
Setting horizon to:NaN
Running WA-STAR
h(n = s_0)=45.0
f(n) = 45.0 (Expanded Nodes: 0, Evaluated States: 0, Time: 0.005) Frontier Size: 0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (alight user bus1 stop-j)

Plan-Length:11
Metric (Search):18.0
Planning Time (msec): 221
Heuristic Time (msec): 2
Search Time (msec): 19
Expanded Nodes:12
States Evaluated:38
```

# Comparison between heuristic **hmax** and all search algorithms on instance 3

```
Domain parsed
Problem parsed
Grounding..
Grounding Time: 22
Aibr Preprocessing
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 8
Setting horizon to:NaN
Running greedy WA-STAR with hw = 4
h(n = s_0)=18.0
f(n) = 72.0 (Expanded Nodes: 0, Evaluated States: 0, Time: 0.006) Frontier Size: 0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (alight user bus1 stop-j)

Plan-Length:11
Metric (Search):18.0
Planning Time (msec): 221
Heuristic Time (msec): 6
Search Time (msec): 24
Expanded Nodes:17
States Evaluated:61
```

```
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 9
Setting horizon to:NaN
Running Greedy Best First Search
h(n = s_0)=18.0
 g(n)= 1.0 h(n)=17.0
 g(n)= 2.0 h(n)=16.0
 g(n)= 9.0 h(n)=11.0
 g(n)= 14.0 h(n)=4.0
 g(n)= 18.0 h(n)=0.0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (move-vehicle tram2 stop-a stop-d)
11.0: (alight user bus1 stop-j)

Plan-Length:12
Metric (Search):25.0
Planning Time (msec): 212
Heuristic Time (msec): 9
Search Time (msec): 20
Expanded Nodes:24
States Evaluated:99
```

```
Domain parsed
Problem parsed
Grounding..
Grounding Time: 19
Aibr Preprocessing
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 10
Setting horizon to:NaN
Running WA-STAR
h(n = s_0)=18.0
f(n) = 18.0 (Expanded Nodes: 0, Evaluated States: 0, Time: 0.008) Frontier Size: 0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (alight user bus1 stop-j)

Plan-Length:11
Metric (Search):18.0
Planning Time (msec): 221
Heuristic Time (msec): 6
Search Time (msec): 22
Expanded Nodes:12
States Evaluated:38
```

# Comparison between heuristic **hradd** and all search algorithms on instance 3

```
Domain parsed
Problem parsed
Grounding..
Grounding Time: 23
Aibr Preprocessing
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 12
Setting horizon to:NaN
Running greedy WA-STAR with hw = 4
h(n = s_0)=45.0
f(n) = 180.0 (Expanded Nodes: 0, Evaluated States: 0, Time: 0.007) Frontier Size: 0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (alight user bus1 stop-j)

Plan-Length:11
Metric (Search):18.0
Planning Time (msec): 237
Heuristic Time (msec): 8
Search Time (msec): 23
Expanded Nodes:12
States Evaluated:38
```

```
Grounding Time: 19
Aibr Preprocessing
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 9
Setting horizon to:NaN
Running Greedy Best First Search
h(n = s_0)=45.0
 g(n)= 1.0 h(n)=40.0
 g(n)= 2.0 h(n)=35.0
 g(n)= 5.0 h(n)=23.0
 g(n)= 6.0 h(n)=20.0
 g(n)= 7.0 h(n)=17.0
 g(n)= 14.0 h(n)=4.0
 g(n)= 18.0 h(n)=0.0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (alight user bus1 stop-j)

Plan-Length:11
Metric (Search):18.0
Planning Time (msec): 222
Heuristic Time (msec): 7
Search Time (msec): 18
Expanded Nodes:13
States Evaluated:44
```

```
Domain parsed
Problem parsed
Grounding..
Grounding Time: 22
Aibr Preprocessing
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 9
Setting horizon to:NaN
Running WA-STAR
h(n = s_0)=45.0
f(n) = 45.0 (Expanded Nodes: 0, Evaluated States: 0, Time: 0.007) Frontier Size: 0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (alight user bus1 stop-j)

Plan-Length:11
Metric (Search):18.0
Planning Time (msec): 223
Heuristic Time (msec): 7
Search Time (msec): 23
Expanded Nodes:12
States Evaluated:38
```

# Comparison between heuristic **hrmax** and all search algorithms on instance 3

```
Domain parsed
Problem parsed
Grounding..
Grounding Time: 21
Aibr Preprocessing
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 10
Setting horizon to:NaN
Running greedy WA-STAR with hw = 4
h(n = s_0)=18.0
f(n) = 72.0 (Expanded Nodes: 0, Evaluated States: 0, Time: 0.006) Frontier Size: 0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (alight user bus1 stop-j)

Plan-Length:11
Metric (Search):18.0
Planning Time (msec): 261
Heuristic Time (msec): 8
Search Time (msec): 30
Expanded Nodes:17
States Evaluated:61
```

```
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 11
Setting horizon to:NaN
Running Greedy Best First Search
h(n = s_0)=18.0
 g(n)= 1.0 h(n)=17.0
 g(n)= 2.0 h(n)=16.0
 g(n)= 9.0 h(n)=11.0
 g(n)= 14.0 h(n)=4.0
 g(n)= 18.0 h(n)=0.0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (move-vehicle tram2 stop-a stop-d)
11.0: (alight user bus1 stop-j)

Plan-Length:12
Metric (Search):25.0
Planning Time (msec): 221
Heuristic Time (msec): 5
Search Time (msec): 21
Expanded Nodes:24
States Evaluated:99
```

```
Domain parsed
Problem parsed
Grounding..
Grounding Time: 19
Aibr Preprocessing
|F|:36
|X|:1
|A|:54
|P|:0
|E|:0
H1 Setup Time (msec): 9
Setting horizon to:NaN
Running WA-STAR
h(n = s_0)=18.0
f(n) = 18.0 (Expanded Nodes: 0, Evaluated States: 0, Time: 0.009) Frontier Size: 0
Problem Solved

Found Plan:
0.0: (wait user stop-a)
1.0: (wait user stop-a)
2.0: (board user tram1 stop-a)
3.0: (move-vehicle tram1 stop-a stop-b)
4.0: (alight user tram1 stop-b)
5.0: (wait user stop-b)
6.0: (wait user stop-b)
7.0: (board user bus1 stop-b)
8.0: (move-vehicle bus1 stop-b stop-f)
9.0: (move-vehicle bus1 stop-f stop-j)
10.0: (alight user bus1 stop-j)

Plan-Length:11
Metric (Search):18.0
Planning Time (msec): 213
Heuristic Time (msec): 3
Search Time (msec): 22
Expanded Nodes:12
States Evaluated:38
```

# Reasoning with IndiGolog

Enable intelligent decision-making in autonomous agents through high-level logical reasoning about actions, events, and their outcomes.

**Use case in this project:**
Calculate the fastest route from point A to B using available connections, time schedules, and transport types—leveraging Prolog rules and IndiGolog reasoning.

SWI Prolog

# Reasoning task 1

**Goal**:

Find the fastest route from a start to a destination using available transport options.

**Inputs**:

- Facts about transport types, routes, durations
- Start time and location

**Output**:

- Step-by-step route plan with timestamps

# Output Reasoning Task 1

Goal:
- going from a to c

Starting travel:
- 8:00

Vehicles:
- autobus1, autobus2, autobus3, autobus4, metro1

```
?- solve_routing_task(a, c, ora(8,0)).

=== ROUTING REASONING TASK ===
Question: What is the fastest way to travel from a to c?
Departure time: ora(8,0)

Execution: System analyzed all available routes and schedules

All possible routes:
  [a,b,c] via autobus1 + metro1, arrival: ora(8,30), duration: 30 min
  [a,b,c] via autobus1 + autobus3, arrival: ora(9,5), duration: 65 min
  [a,b,c] via autobus2 + autobus3, arrival: ora(9,5), duration: 65 min
  [a,c] via autobus4, arrival: ora(9,25), duration: 85 min

Answer: The fastest route is [a,b,c]
        Transportation: autobus1 + metro1
        Departure: ora(8,0)
        Arrival: ora(8,30)
        Total travel time: 30 minutes
true .
```

# Reasoning task 2

**Task Description**:

What happens if one of the transports is delayed?

**Objective**:

- Analyze the original (fastest) route from a given departure point and time.
- Apply a delay to a specific vehicle.
- Determine if alternative routes become better options.

# Output Reasoning Task 2

**Goal:**
- going from a to c

**Starting travel:**
- 8:00

**Vehicles delayed:**
- autobus1

```
?-  solve_delay_task(a, c, ora(8,0), autobus1, 20).

=== DELAY HANDLING REASONING TASK ===
Question: What are the travel options if transport autobus1 is delayed by 20 minutes?
Route: a to c, planned departure after ora(8,0)

Execution: Analyzing all routes before and after the delay.

--- 1. AVAILABLE ROUTES (NO DELAYS) ---
Found 4 possible original route(s):
  - Route: [a,b,c] via autobus1 -> autobus2
    Arrival: ora(8,55), Duration: 55 min
  - Route: [a,b,c] via autobus1 -> metro2
    Arrival: ora(9,12), Duration: 72 min
  - Route: [a,b,c] via metro1 -> metro2
    Arrival: ora(9,12), Duration: 72 min
  - Route: [a,c] via treno_diretto
    Arrival: ora(9,50), Duration: 110 min

--[LOG: Aggiunto ritardo di 20 minuti per autobus1]--
--- 2. ALTERNATIVE ROUTES (WITH DELAY) ---
Found 3 possible route(s) considering the delay:
  - Route: [a,b,c] via autobus1 -> metro2
    Arrival: ora(9,12), Duration: 72 min (Delay info: autobus1 delayed by 20 min)
  - Route: [a,b,c] via metro1 -> metro2
    Arrival: ora(9,12), Duration: 72 min (Delay info: no delays)
  - Route: [a,c] via treno_diretto
    Arrival: ora(9,50), Duration: 110 min (Delay info: no delays)

--- ANALYSIS OF UNAVAILABLE ORIGINAL ROUTES ---
  - NOTE: The original route [a,b,c] via autobus1 -> autobus2 is NO LONGER FEASIBLE due to a missed connection.
All original routes were checked.

--- 3. FINAL ANSWER AND SUMMARY ---
The new best option is the route [a,b,c] via autobus1 -> metro2.
  - New estimated arrival: ora(9,12) (Total duration: 72 min).
  - This route is affected by: autobus1 delayed by 20 min.
  - Impact: This is 17 minutes slower than the original best option.
--[LOG: Rimosso ritardo per autobus1]--
true.
```

# Reasoning task 3

**Goal:**

Determine the updated travel options between two points after introducing an avoided vehicle.

**Initial Setup:**

- Departure city, destination, and earliest departure time.
- A vehicle (e.g., metro1) is avoided while computing all possible routes

# Output Reasoning Task 3

**Goal:**

- going from a to c

**Starting travel:**

- 8:00

**Vehicles delayed:**

- autobus1

```
?-  solve_preference_task(a, c, ora(8,0), [autobus1]).

=== PREFERENCE-BASED REASONING TASK ===
Question: Can a user reach the destination while avoiding autobus1?
Route: a to c, planned departure after ora(8,0)

Execution: Finding all routes and filtering based on user preferences.

--- 1. ANALYSIS OF AVAILABLE ROUTES ---
Found 2 route(s) that satisfy the preference:
  - Route: [a,b,c] via metro1 -> metro2
    Arrival: ora(9,12), Duration: 72 min
  - Route: [a,c] via treno_diretto
    Arrival: ora(9,50), Duration: 110 min

--- 2. FINAL ANSWER ---
Answer: Yes, there is a route available that avoids autobus1.
The best option is the route [a,b,c] via metro1 -> metro2.
  - Estimated arrival: ora(9,12) (Total duration: 72 min).
true .
```

# Challenges and limitations

1. Limited support for complex arithmetic
2. Handling dynamic changes
3. Scalability

1. Steep learning curve
2. Limited documentation and debugging
3. Runtime performance

# Thank You!

Agnese Chiessi | Ludovica Moscato